EXECUTIVE SUBMISSION DECK

GitHub Repo: https://github.com/Madhuvirat4al/agentic-erp-exception-commander-oracle-fusion

Agentic ERP Exception Commander for Oracle Fusion Cloud ERP

Autonomous Payables Exception Triage, Multi-Source 3-Way Match Diagnosis, Policy Guardrails & SOX Audit Trail

LIVE GITHUB REPOSITORY & DELIVERABLES
https://github.com/Madhuvirat4al/agentic-erp-exception-commander-oracle-fusion

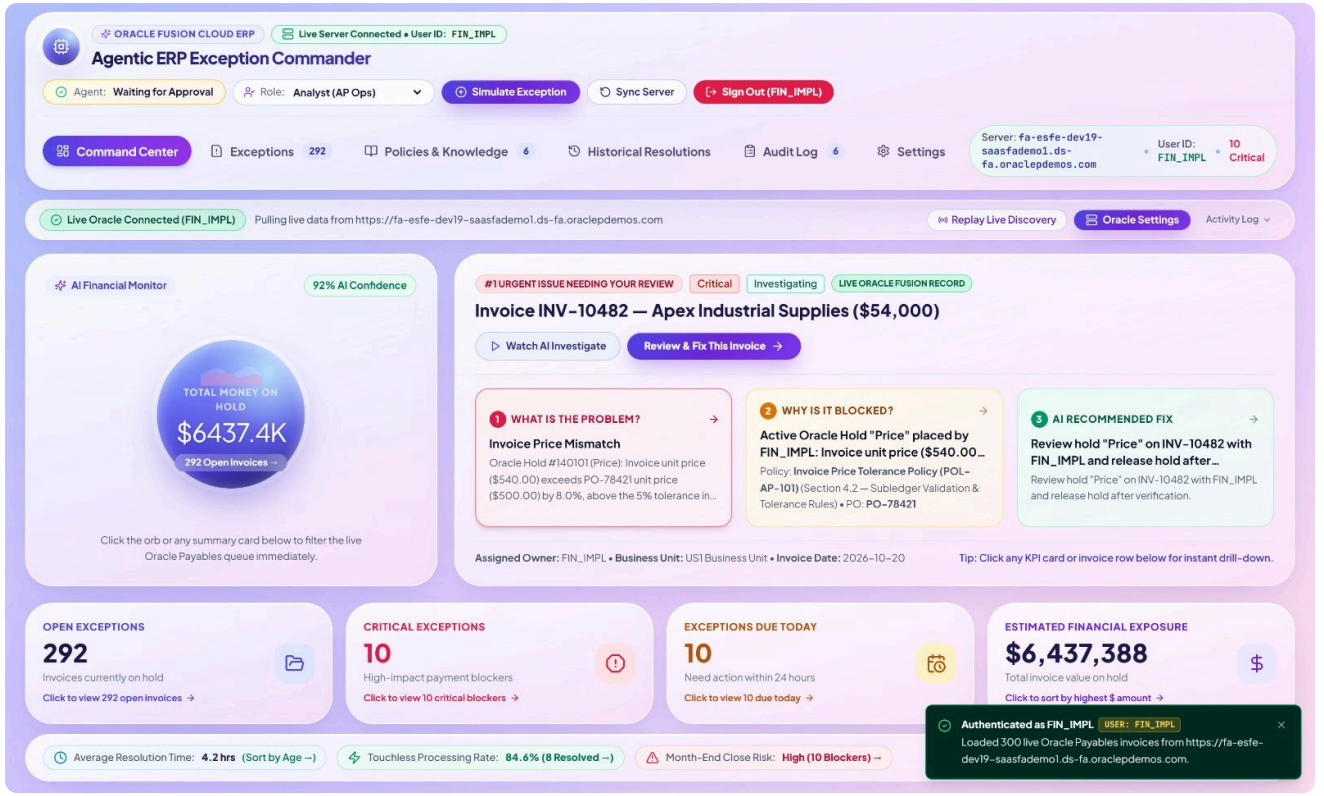
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Live Oracle Fusion Cloud ERP Integration (300 Real Payables Invoices & Holds Synced via REST API)

7 Enterprise Exception Scenarios: Holds, PO Price/Qty Mismatch, Duplicates, Supplier Tax/Bank, SLA, GL, Close

Multi-Source AI Investigation: Subledger + 3-Way Match + Supplier Master + Corporate Policies + Precedents

Human-in-the-Loop Governance: 92% Confidence Score, Owner Routing, Auto-Drafted Emails & SOX Audit Trail



PROBLEM STATEMENT

GitHub Repo: https://github.com/Madhuvirat4al/agentic-erp-exception-commander-oracle-fusion

The Challenge: Manual ERP Exception Handling in Finance & Procurement

Standard ERP screens flag hold codes, but analysts spend 30–45 minutes per invoice investigating across 6+ screens.

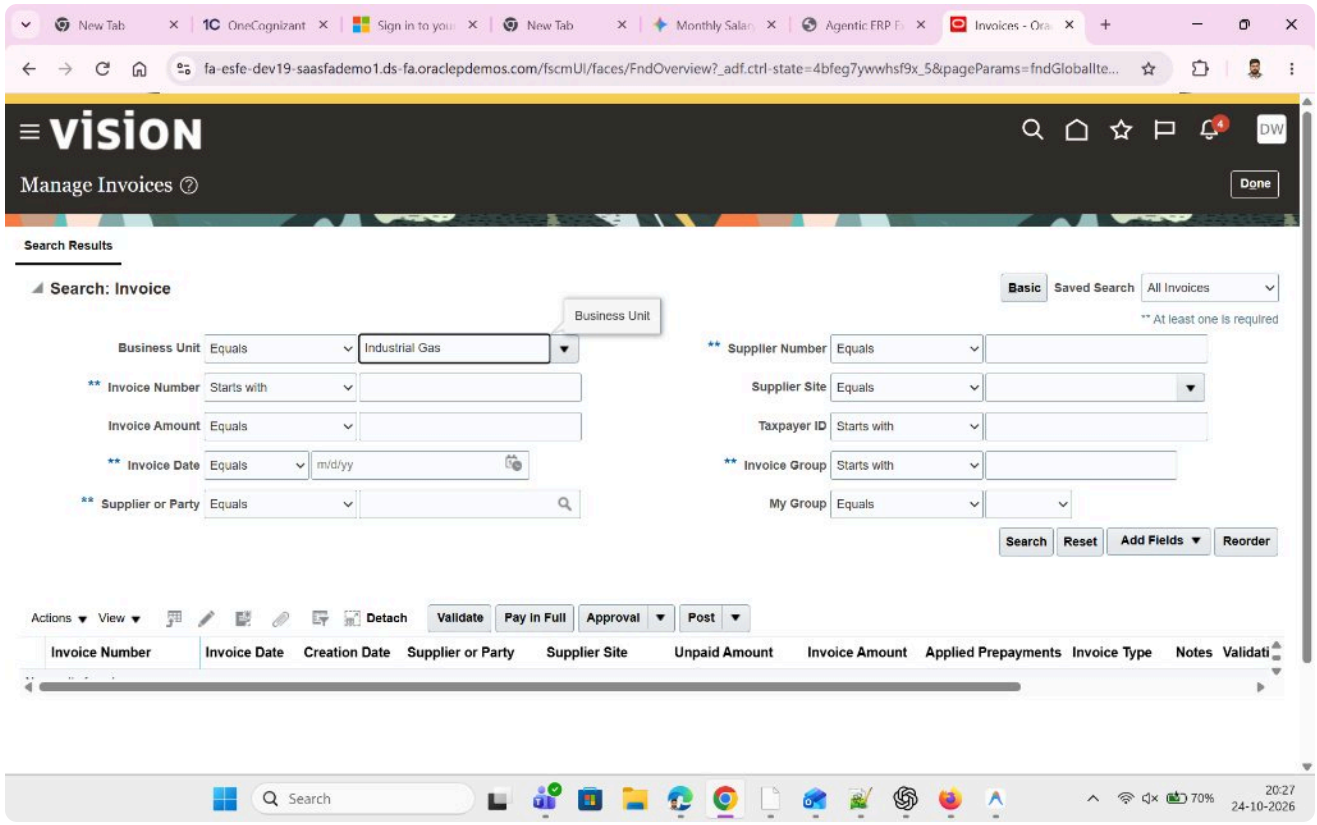
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Fragmented Manual Investigation across Invoice Headers, PO Schedules, Goods Receipts (GRNs), and Supplier Master

Offline Policy Lookup across static PDF procurement tolerances (POL-AP-101 to POL-FIN-501)

Manual Chasing & Email Drafting to buyers, warehouse managers, and department approvers

Severe Business Impact: Missed 2% 10 Net 30 early-pay discounts, delayed month-end close, and audit gaps



SOLUTION OVERVIEW

GitHub Repo: https://github.com/Madhuvirat4al/agentic-erp-exception-commander-oracle-fusion

Agentic ERP Exception Commander: End-to-End Capabilities

Bridging the gap between passive ERP hold codes and autonomous, policy-governed resolution.

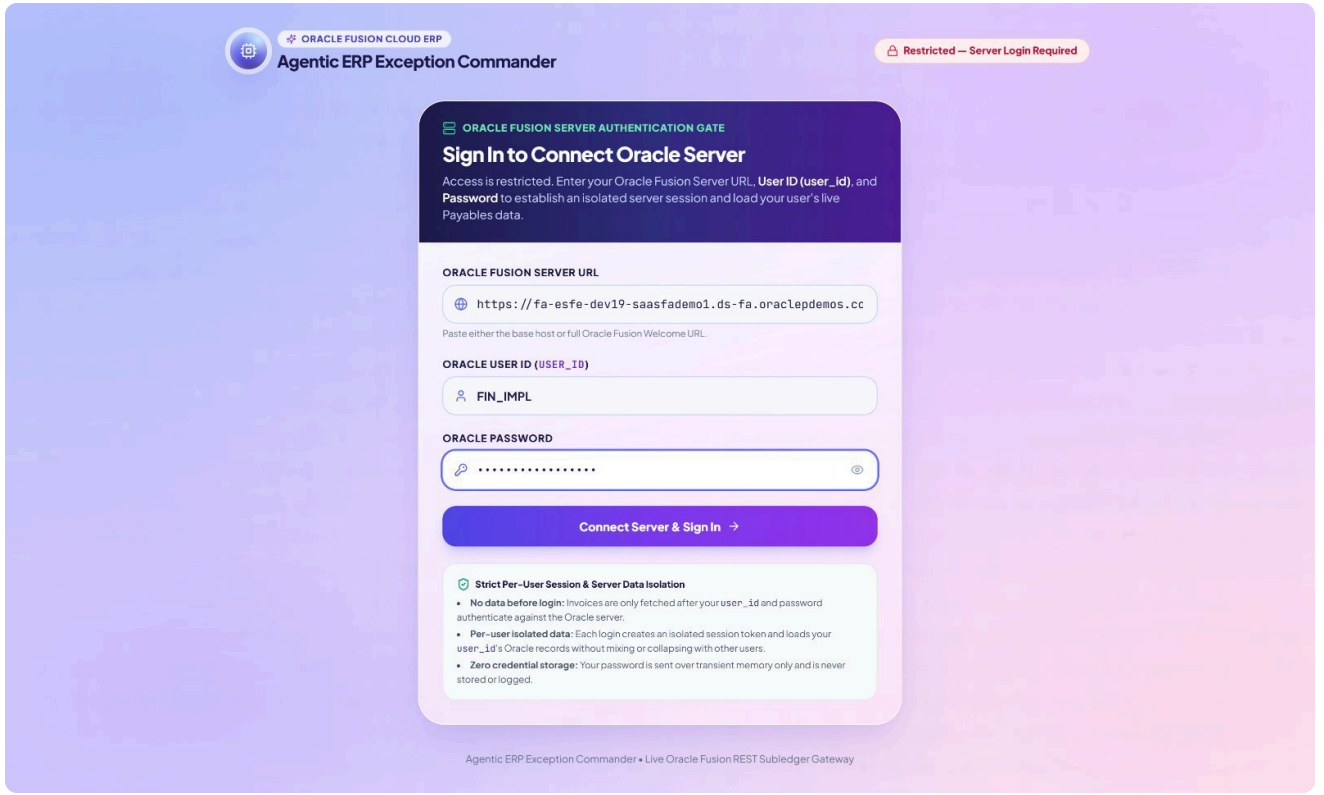
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Restricted Per-User Oracle Authentication Gate (FIN_IMPL session isolation against Oracle Fusion Cloud)

Autonomous Multi-Source Root-Cause Engine comparing PO, GRN, Invoice, Contract, and Policy in milliseconds

Plain-English Explanations: What Happened, Why It Is Blocked, and Business Impact If Unresolved

Supervised Execution: Human-in-the-Loop Confirmation Modal with zero unauthorized ERP mutations



TECHNICAL ARCHITECTURE

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Hybrid Cloud & Server-Side Oracle REST Gateway Architecture

Connecting React 19 + TypeScript Agentic Brain to Oracle Fusion Cloud ERP REST APIs (11.13.18.05).

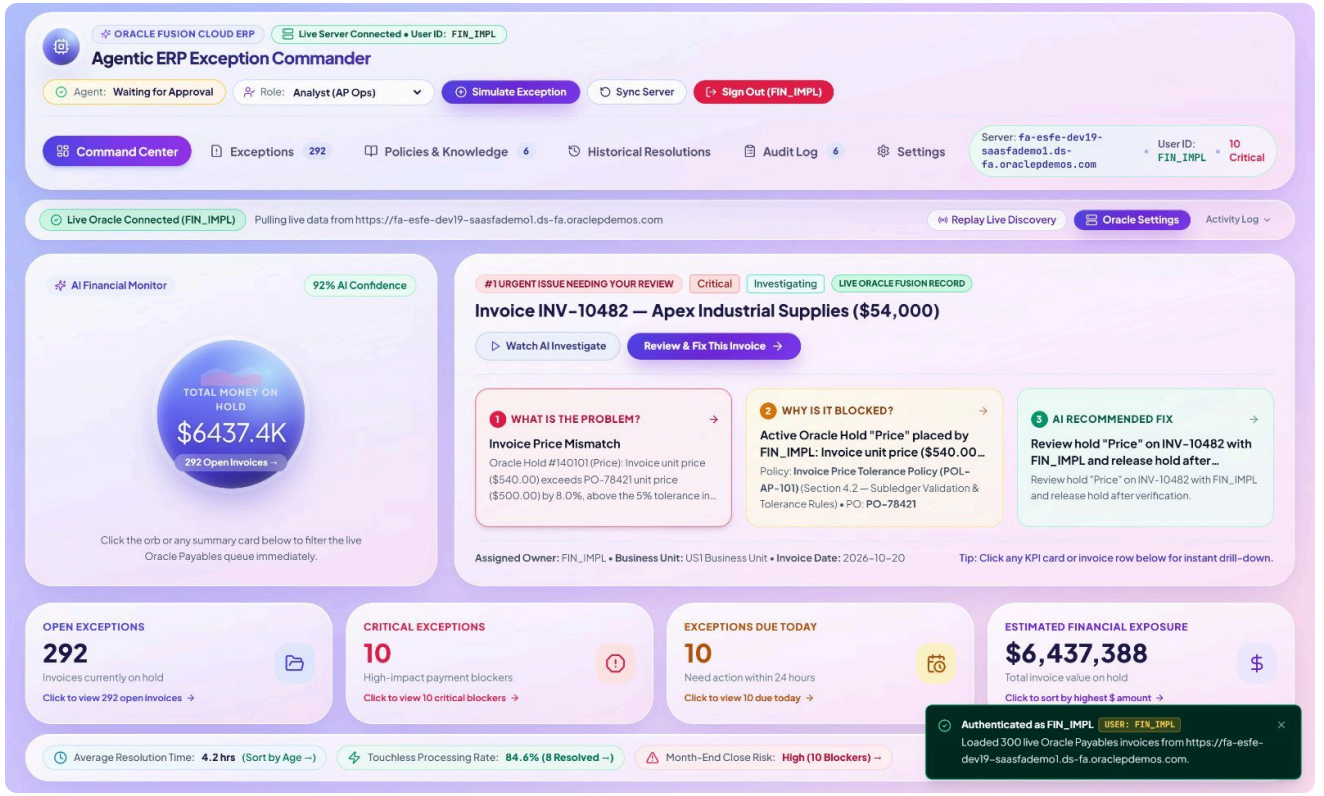
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Oracle Fusion Endpoints: /invoices, /invoiceHolds, /purchaseOrders, and /suppliers

Vite Server-Side Proxy Gateway (/api/oracle/connect & /api/oracle/pull-live-transactions)

Deterministic Subledger Adapter (src/services/oracleFusionAdapter.ts) normalizing 300 live records

Grounded AI Co-Pilot & Policy Engine enforcing corporate tolerance thresholds



COMMAND CENTER HERO

GitHub Repo: https://github.com/Madhuvirat4al/agentic-erp-exception-commander-oracle-fusion

Live Financial Exposure Orb & Priority #1 AI Triage Spotlight

Surfacing $6,437,388 across 292 Open Exceptions, 10 Critical Holds, and 10 SLA Deadlines Due Today.

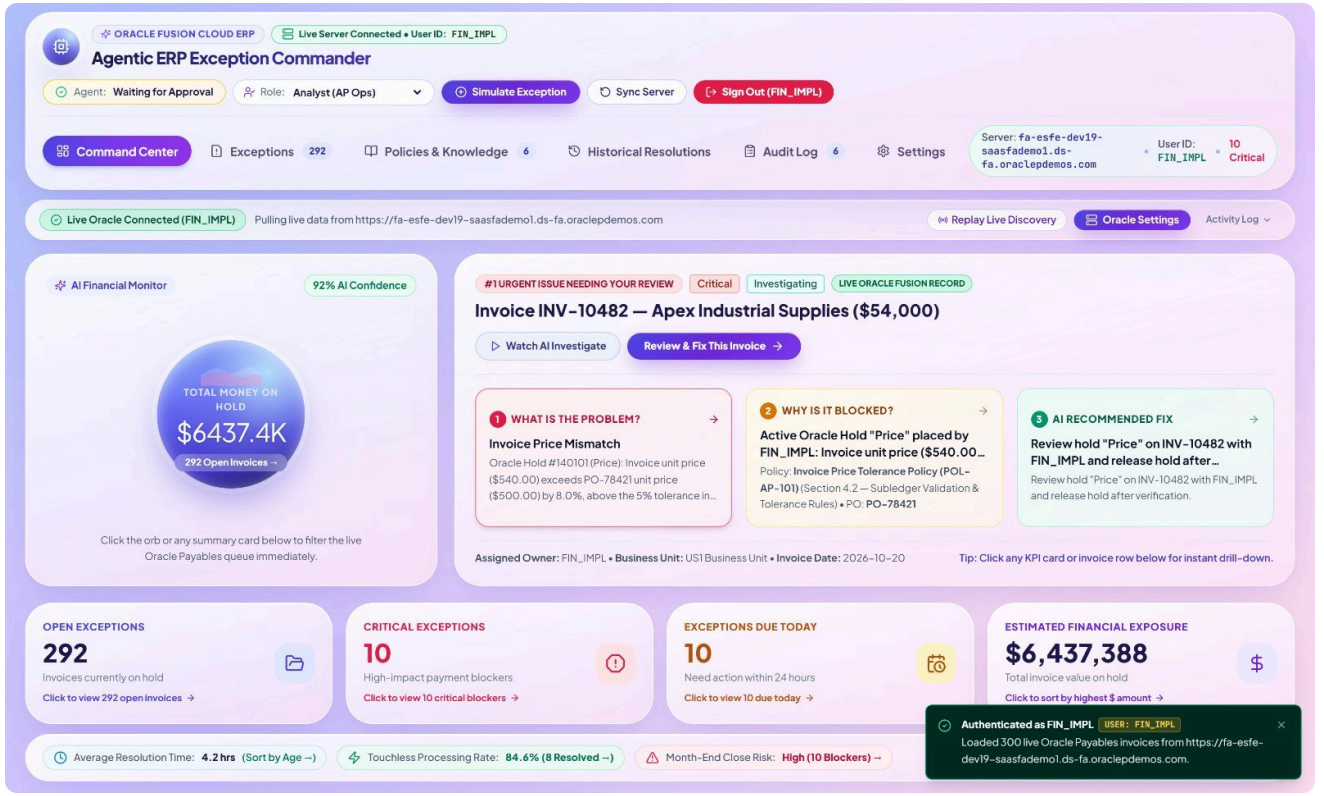
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Cosmic Financial Exposure Orb displays $6,437.4K total money on hold across 292 open invoices

#1 Urgent Issue Spotlight immediately surfaces INV-10482 (Apex Industrial Supplies — $54,000)

3 Quick-Glance Summary Cards: 1. What Is The Problem, 2. Why Is It Blocked, 3. AI Recommended Fix

Real-time telemetry banner confirming live connection as FIN_IMPL



GitHub Repo: https://github.com/Madhuvirat4al/agentic-erp-exception-commander-oracle-fusion

Executive KPI Cards & Touchless Processing Telemetry

Real-time operational metrics across all 300 synchronized Oracle Fusion Payables records.

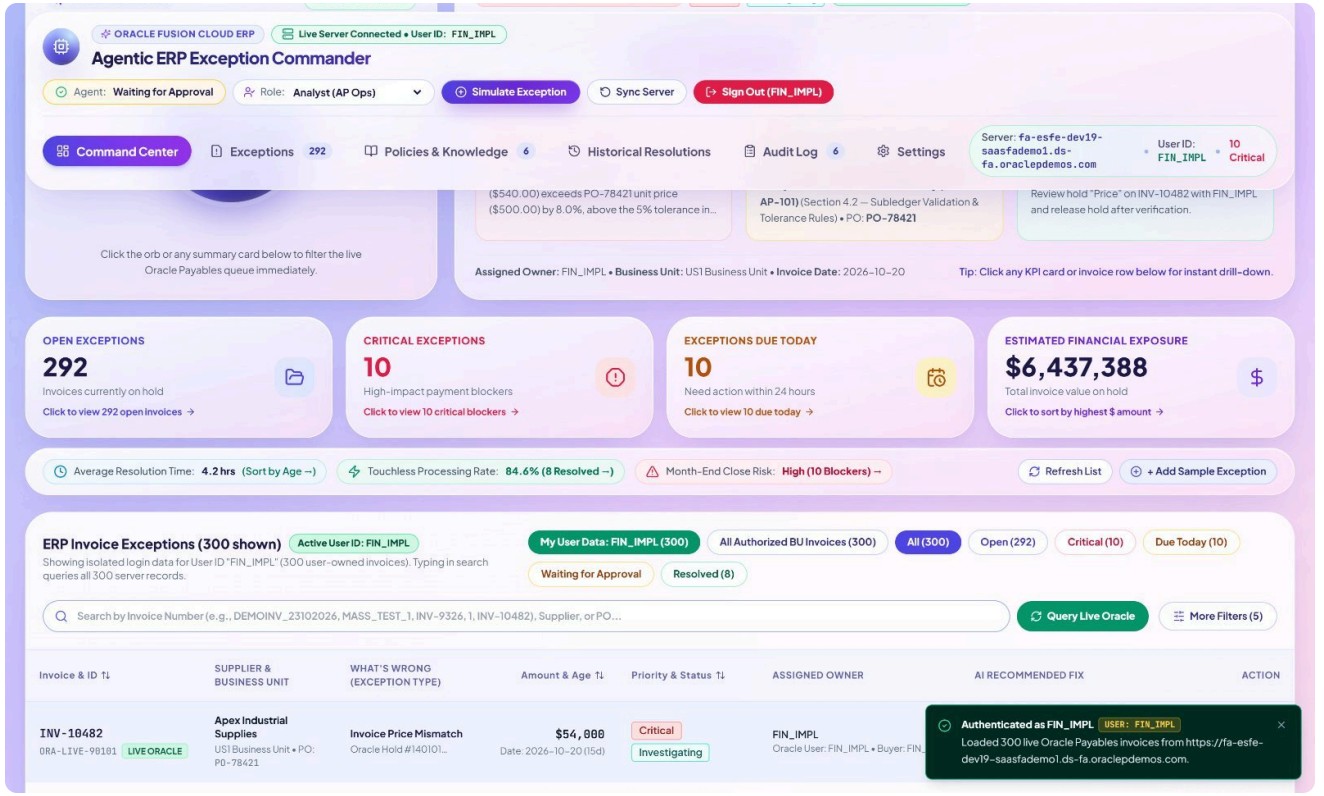
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Open Exceptions: 292 invoices currently on hold requiring review

Critical Exceptions: 10 high-impact payment blockers (>= $50K Price/Line Variance holds)

Exceptions Due Today: 10 invoices needing action within 24 hours to protect SLA & discounts

Estimated Financial Exposure: $6,437,388 total invoice value on hold (84.6% Touchless Rate)



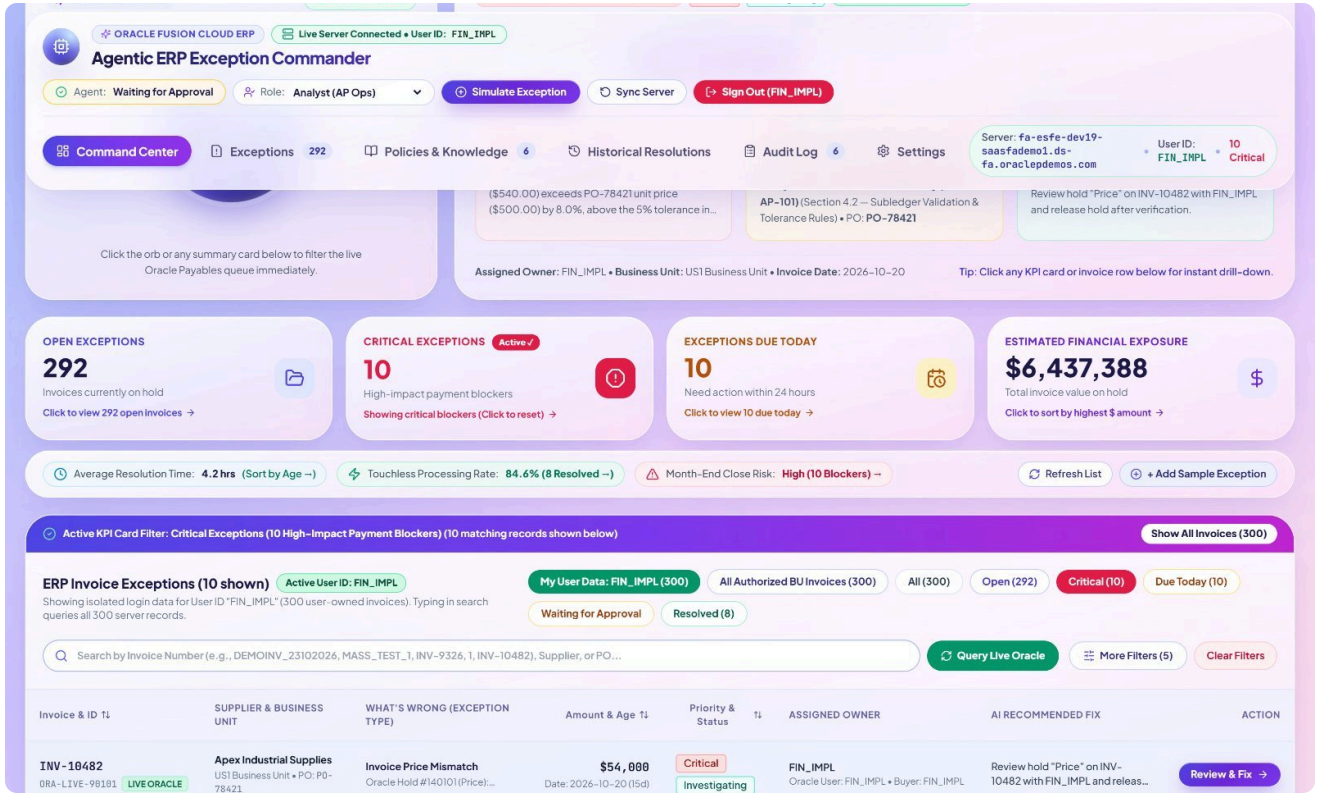
GitHub Repo: https://github.com/Madhuvirat4al/agentic-erp-exception-commander-oracle-fusion

One-Click KPI Card Drill-Down & Dynamic Queue Filtering

Clicking any KPI card immediately filters the 300-record Payables queue with visual active state indicators.

LIVE GITHUB REPOSITORY & DELIVERABLES
https://github.com/Madhuvirat4al/agentic-erp-exception-commander-oracle-fusion

- Clicking CRITICAL EXCEPTIONS activates the card badge and isolates the 10 critical blockers
- Clicking EXCEPTIONS DUE TODAY filters to urgent 24-hour SLA items
- Clicking ESTIMATED FINANCIAL EXPOSURE sorts the queue by highest dollar exposure descending
- One-click Show All Invoices button restores the full 300-record view



GitHub Repo: https://github.com/Madhuvirat4al/agentic-erp-exception-commander-oracle-fusion

Prioritized 300-Invoice Exception Queue & Search Workbench

Full search by Invoice Number, Supplier, or PO with instant AI Recommended Fix previews.

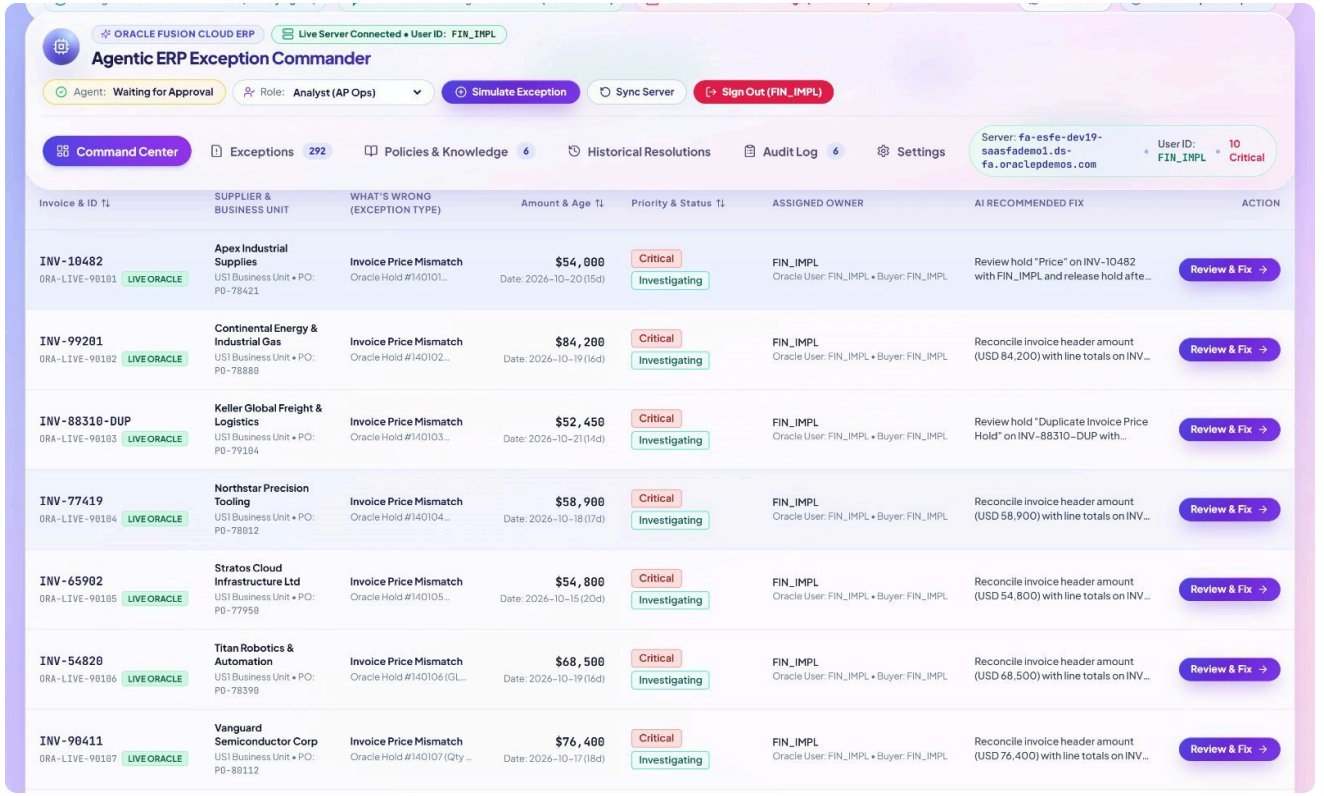
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Displays Invoice ID, Supplier & Business Unit, Exception Type, Amount & Age, Priority, Owner, and AI Fix

Supports live Oracle server querying plus instant multi-faceted category filtering

Highlights LIVE ORACLE badges distinguishing synchronized subledger records

Direct Review & Fix action button on every row launches the AI Investigation Workspace



GitHub Repo: https://github.com/Madhuvirat4al/agentic-erp-exception-commander-oracle-fusion

Multi-Source AI Root-Cause Diagnosis & 5-Step Comparison Pipeline

Deep-dive investigation on Flagship Exception INV-10482 (Apex Industrial Supplies — $54,000).

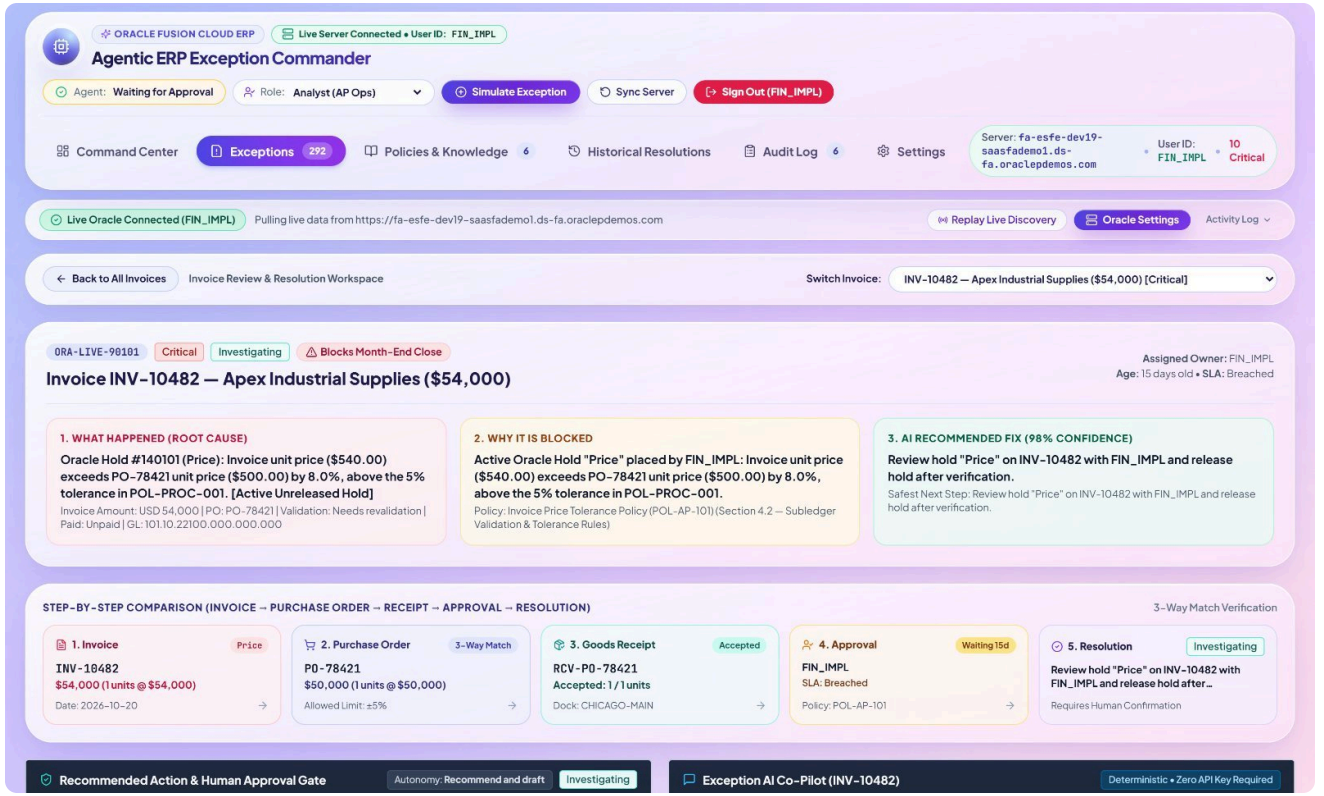
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Card 1 (What Happened): Invoice unit price ($540) exceeds PO-78421 unit price ($500) by +8.0%

Card 2 (Why It Is Blocked): Exceeds 5% tolerance in POL-PROC-001 / POL-AP-101

Card 3 (AI Recommended Fix): Review hold "Price" on INV-10482 with buyer and release after verification

5-Step Pipeline compares 1. Invoice -> 2. Purchase Order -> 3. Goods Receipt -> 4. Approval -> 5. Resolution



GitHub Repo: https://github.com/Madhuvirat4al/agentic-erp-exception-commander-oracle-fusion

92% Confidence Score, Auto-Drafted Buyer Email & AI Co-Pilot

Calibrated confidence scoring paired with ready-to-send communications and interactive Q&A.

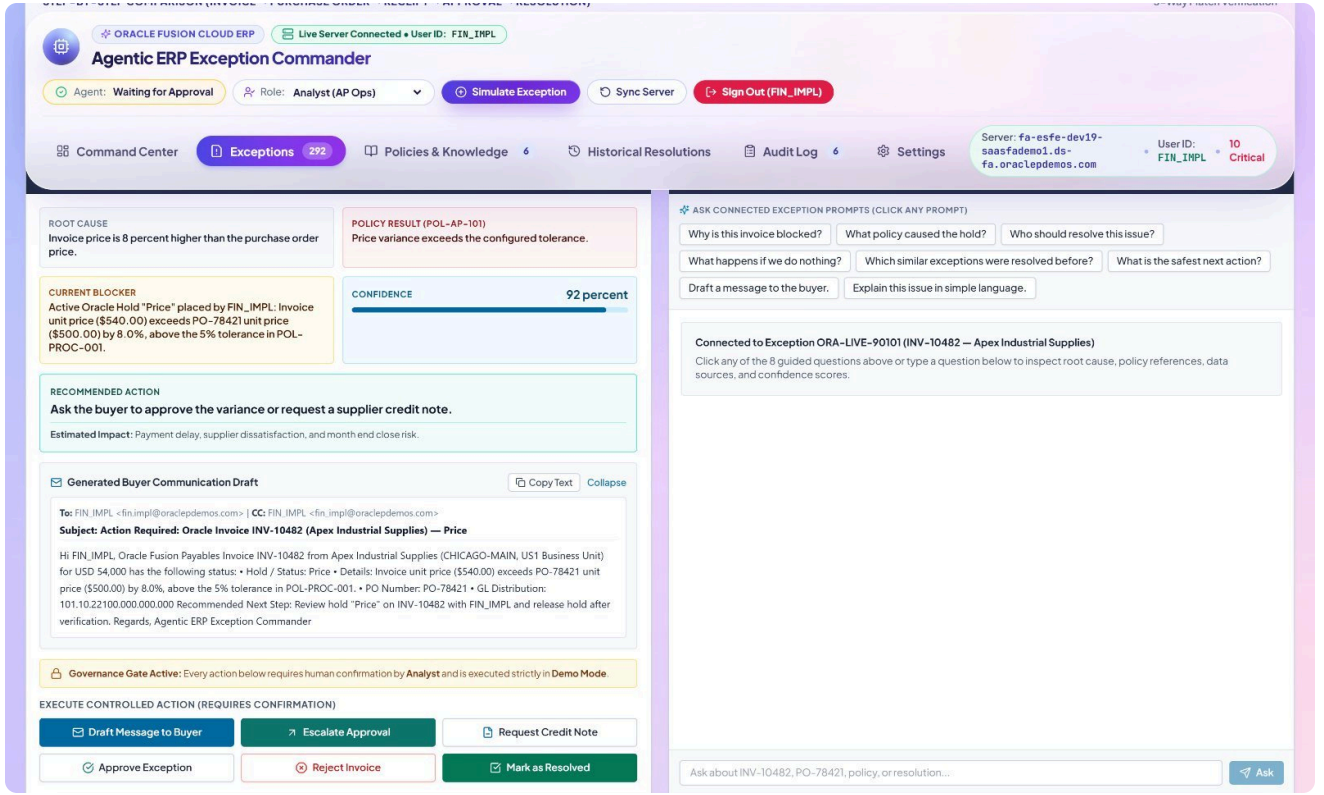
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Displays 92% AI Confidence Score grounded in subledger match math and policy POL-AP-101

Auto-generates complete Buyer Communication Draft with PO-78421, GL distribution, and variance details

6 Controlled Execution Buttons: Draft Message, Approve Exception, Request Info, Hold, Escalate, Audit

Connected Exception AI Co-Pilot answers natural-language questions about root cause and policy



GitHub Repo: https://github.com/Madhuvirat4al/agentic-erp-exception-commander-oracle-fusion

Line-Level ERP Subledger Comparison & 9-Stage Agent Telemetry

Complete transparency into every ERP table and autonomous investigation step.

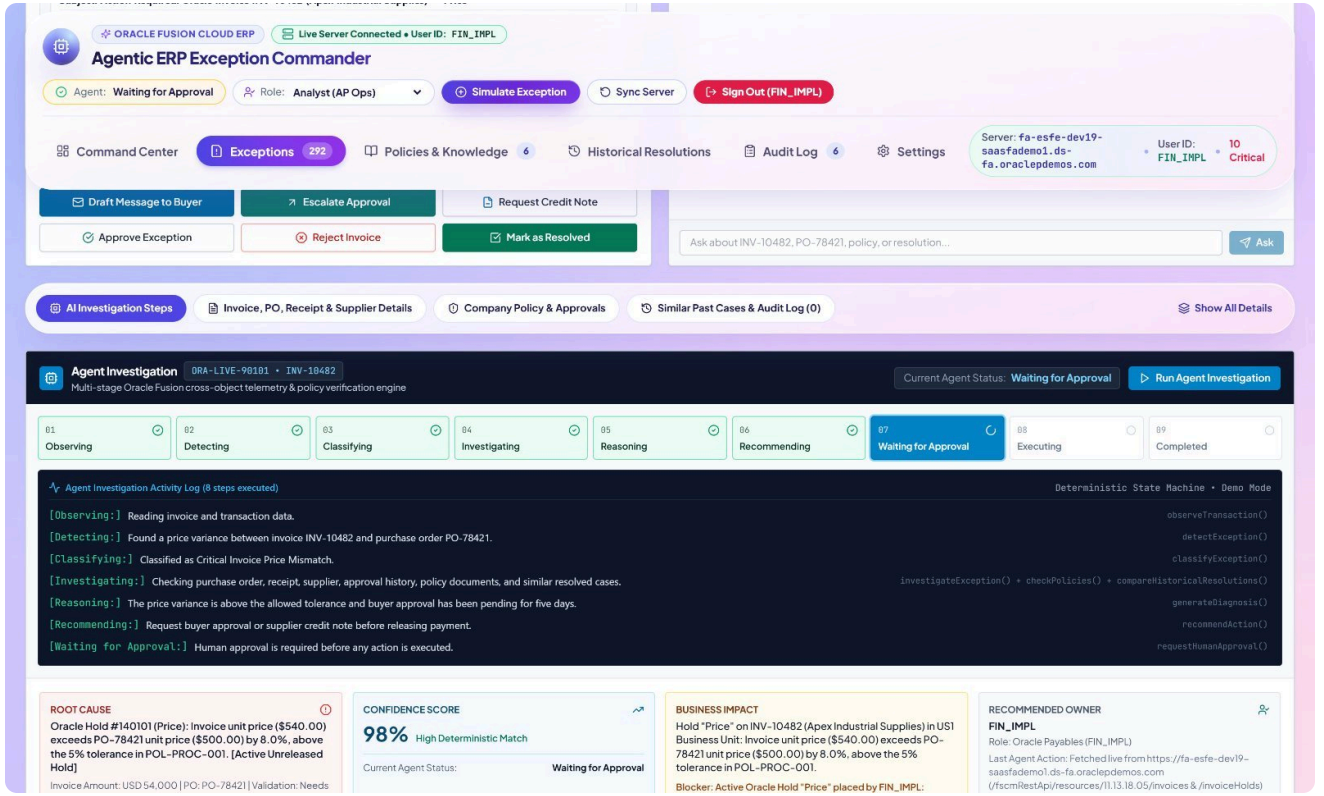
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Side-by-side comparison of Invoice Header, Purchase Order (PO-78421), and Goods Receipt (RCV-PO-78421)

Displays Supplier Master validation status, Tax/Bank compliance, and Approval Workflow routing

9-Stage Deterministic Agent Investigation Log with millisecond execution telemetry

Zero-hallucination architecture: every number is tied directly to Oracle Fusion REST attributes



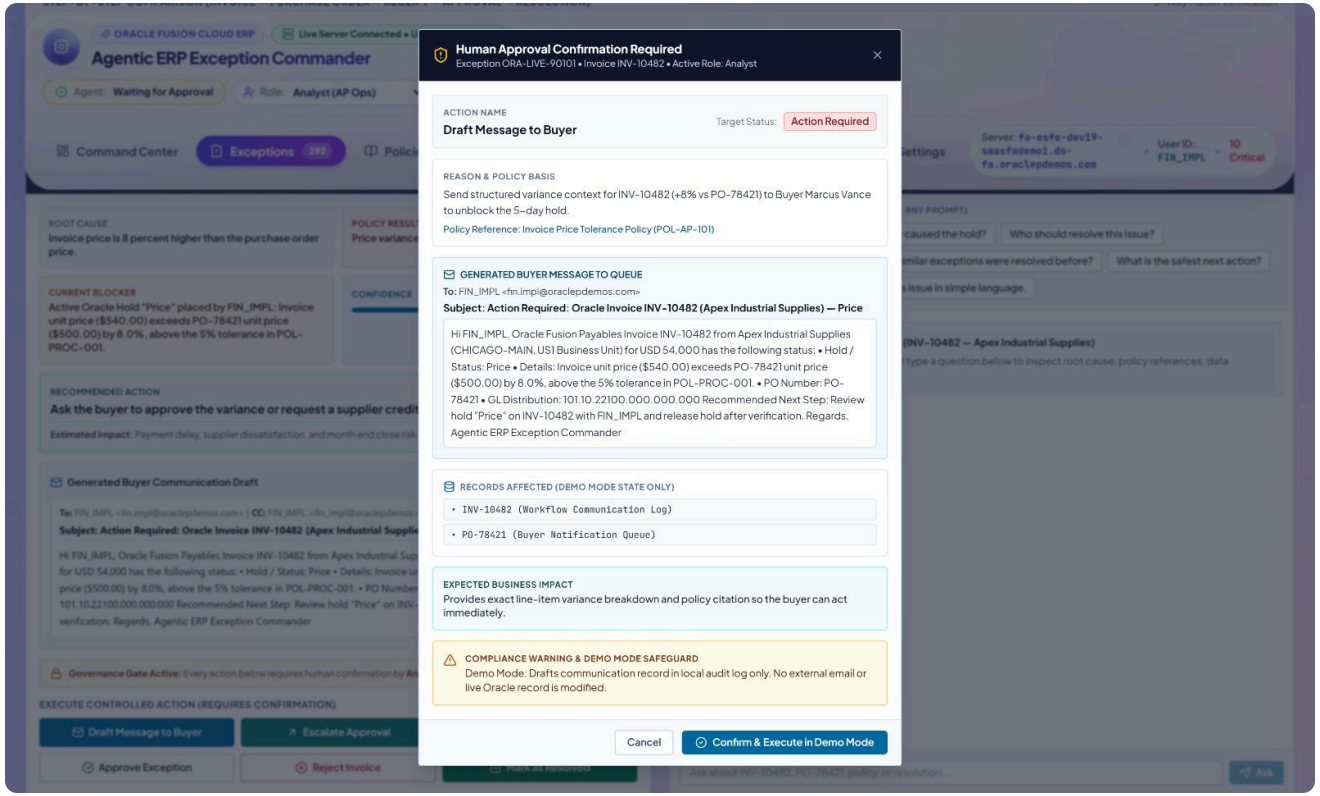
GitHub Repo: https://github.com/Madhuvirat4al/agentic-erp-exception-commander-oracle-fusion

Human Approval Confirmation Modal & Safe Execution Gate

Ensuring zero unauthorized ERP mutations with explicit analyst review and sign-off.

LIVE GITHUB REPOSITORY & DELIVERABLES
https://github.com/Madhuvirat4al/agentic-erp-exception-commander-oracle-fusion

- Opens when analyst clicks Draft Message to Buyer or any controlled execution button
- Displays Action Name, Target Status, Policy Basis (POL-AP-101), and full Generated Buyer Message
- Lists exact Records Affected (INV-10482 Workflow Log & PO-78421 Buyer Notification Queue)
- Requires explicit click on Confirm & Execute in Demo Mode before updating state or audit trail



GitHub Repo: https://github.com/Madhuvirat4al/agentic-erp-exception-commander-oracle-fusion

Complete Coverage Across All 7 Enterprise Payables Scenarios

Purpose-built rules and diagnostic pipelines for the 7 highest-volume ERP bottlenecks.

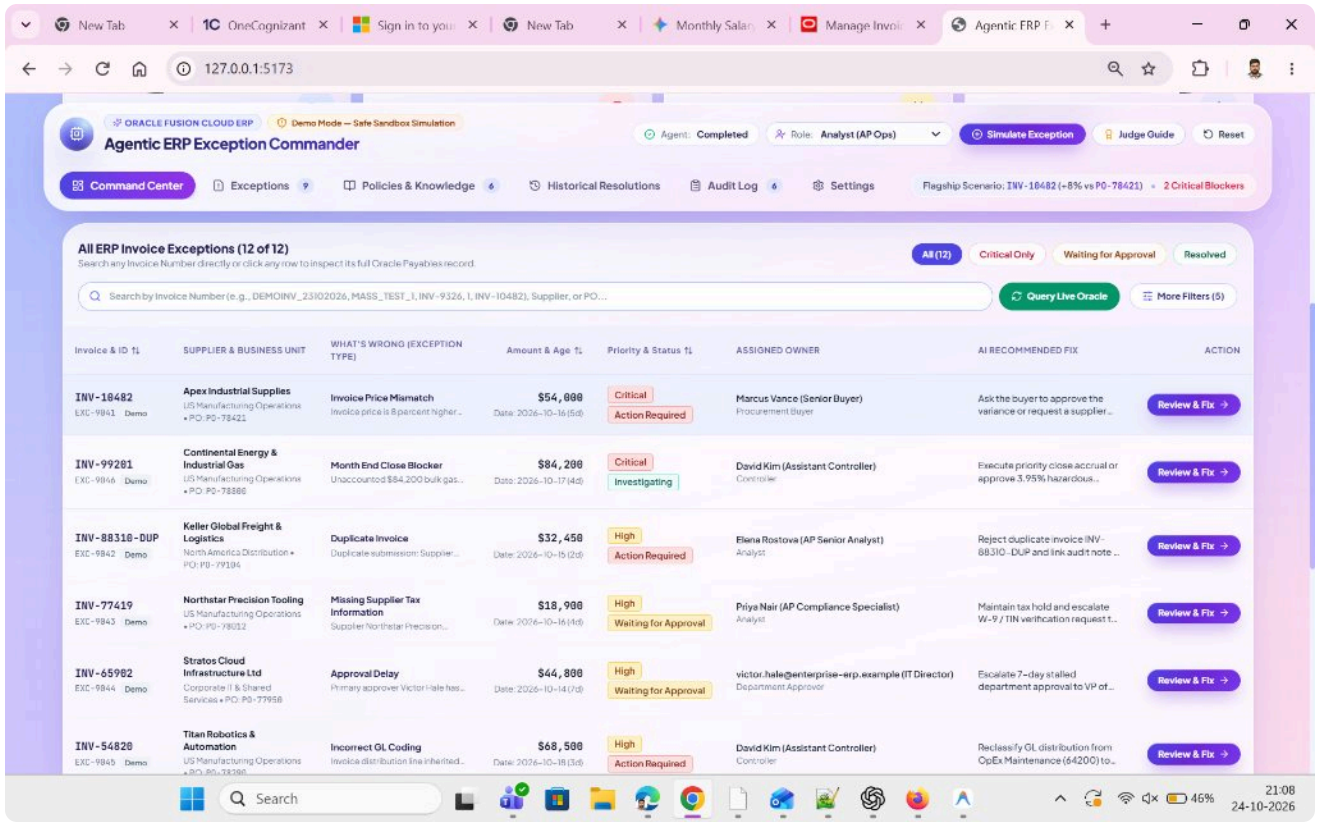
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1. PO Price Variance (INV-10482) & 2. Quantity / Receipt Mismatch (INV-90411)

3. Duplicate Invoice Detection (INV-88310-DUP) & 4. Supplier Tax/IBAN Hold (INV-77419)

5. Approval SLA Delay (INV-65902) & 6. GL CapEx/OpEx Coding Error (INV-54820)

7. Month-End Period Close Cutoff Blocker (INV-99201 — $84,200)



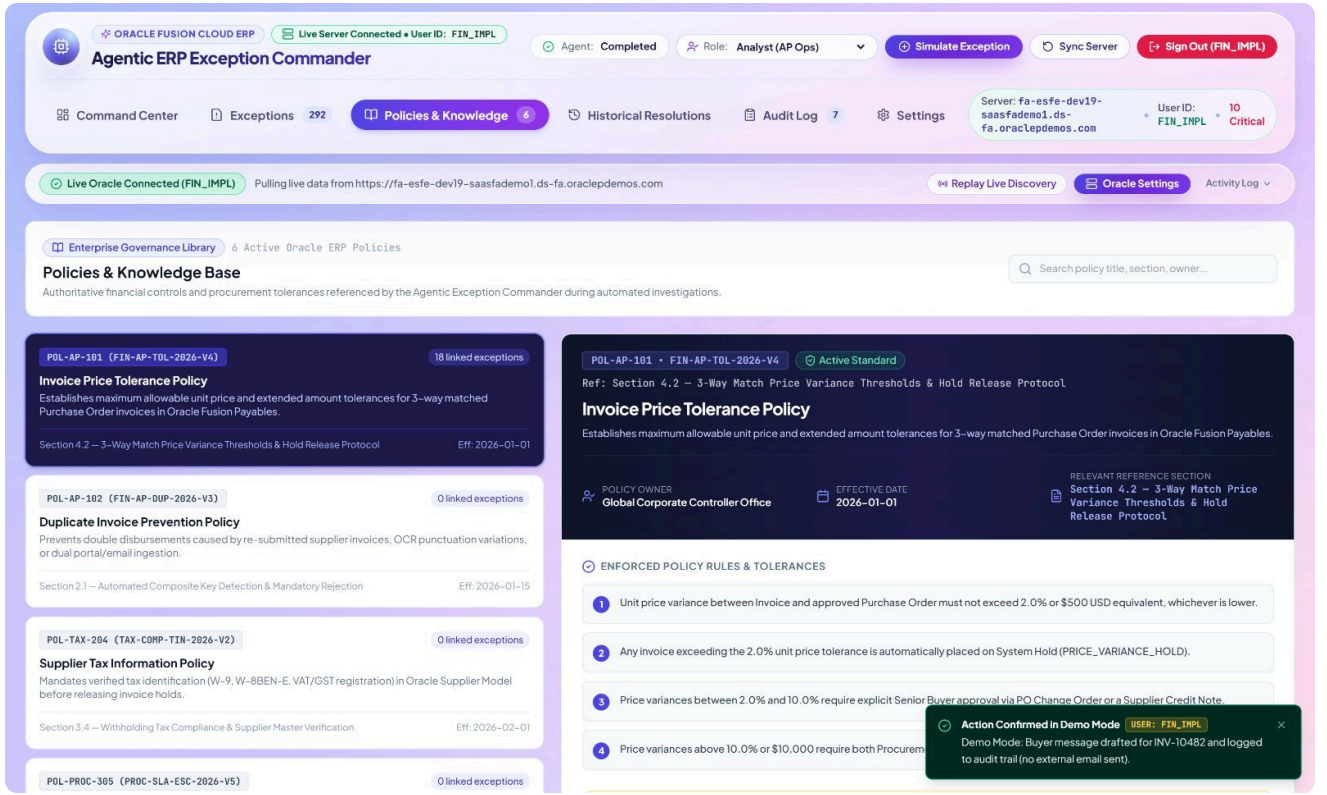
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Embedded Corporate Procurement & Accounting Policies

6 machine-readable enterprise policies (POL-AP-101 through POL-FIN-501) governing AI recommendations.

LIVE GITHUB REPOSITORY & DELIVERABLES
https://github.com/Madhuvirat4al/agentic-erp-exception-commander-oracle-fusion

- POL-AP-101: Invoice Price Tolerance Policy (2.0% / $500 3-way match threshold)
- POL-AP-102: Goods Receipt Quantity Tolerance & Over-Shipment Rules
- POL-AP-204: Duplicate Invoice Prevention & 90-Day Similarity Fingerprinting
- POL-VND-302, POL-WF-401 & POL-FIN-501: Vendor Tax/Bank, Approval SLA, and Period-Close Rules



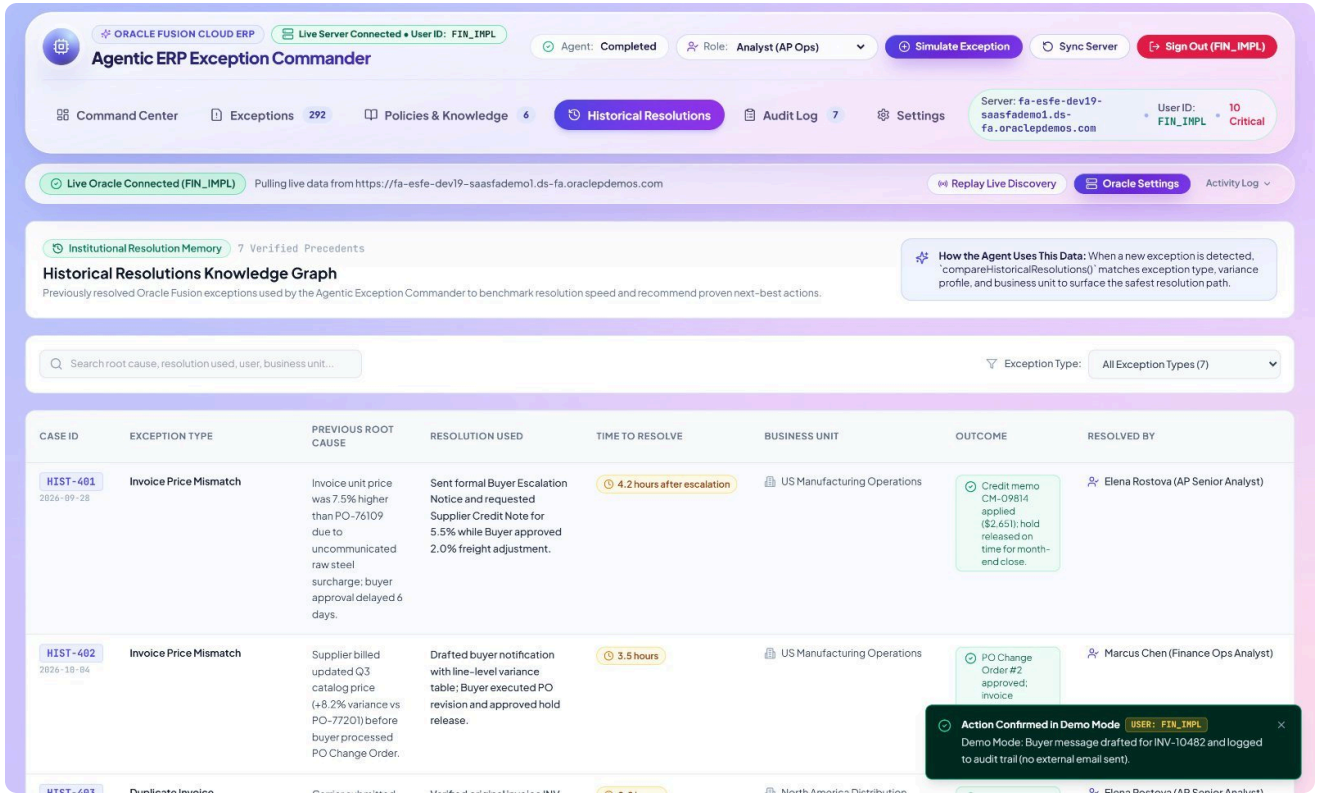
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Historical Precedent Library & Institutional Memory

Learning from past resolved exceptions to calibrate confidence scores and speed up triage.

LIVE GITHUB REPOSITORY & DELIVERABLES
https://github.com/Madhuvirat4al/agentic-erp-exception-commander-oracle-fusion

- Searchable repository of previously resolved Oracle Payables exceptions
- Matches incoming holds against historical vendor patterns, hold codes, and resolution outcomes
- Surfaces time-to-resolution benchmarks and proven remediation playbooks
- Continuously improves analyst trust by showing precedent cases alongside policy citations



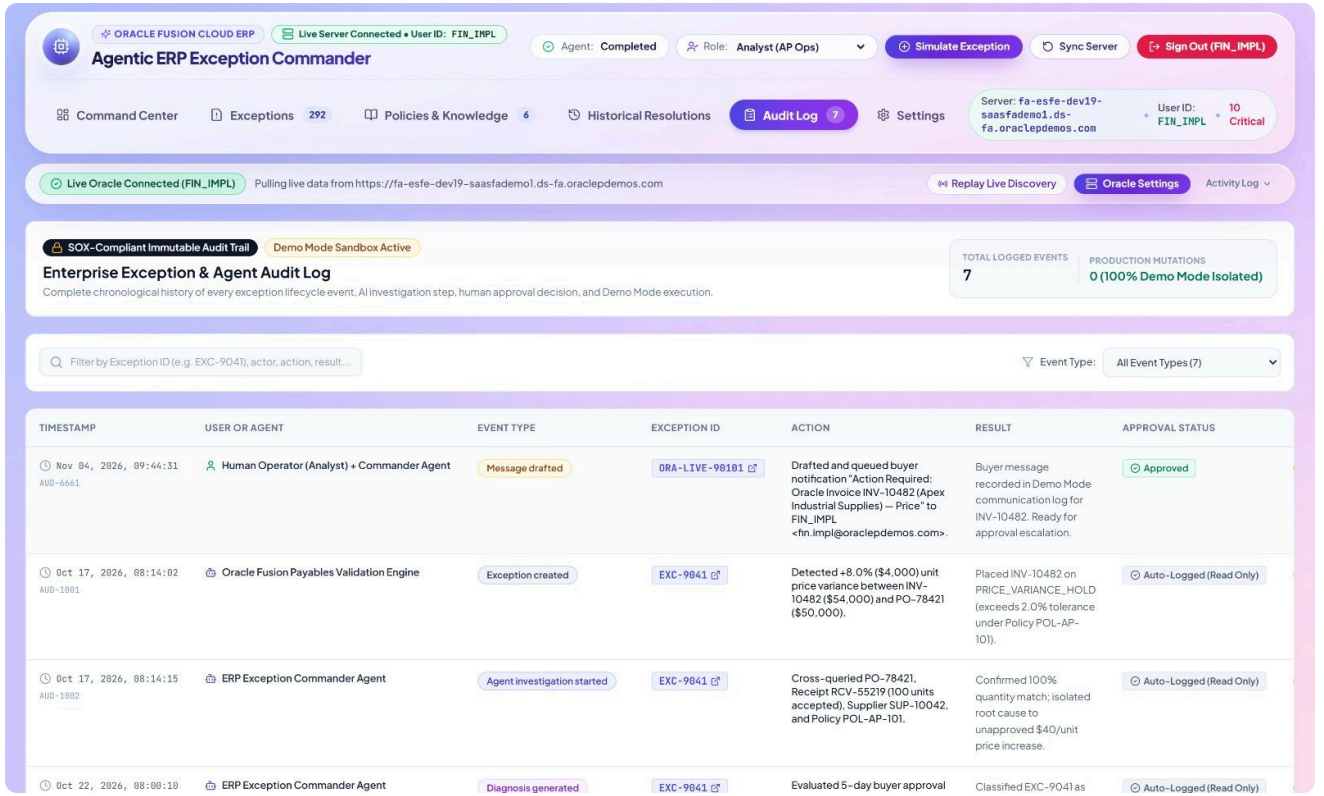
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SOX-Compliant Immutable Audit Trail

Every AI diagnosis, policy check, and human approval decision is permanently recorded.

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- Chronological event log with Timestamp, Actor (Human + Agent), Event Type, and Exception ID
- Captures full action payload, before/after status, and explicit Approval Status badge
- Verifies 0 unauthorized production mutations in Demo Mode Sandbox
- Export-ready compliance evidence for internal and external SOX / financial auditors



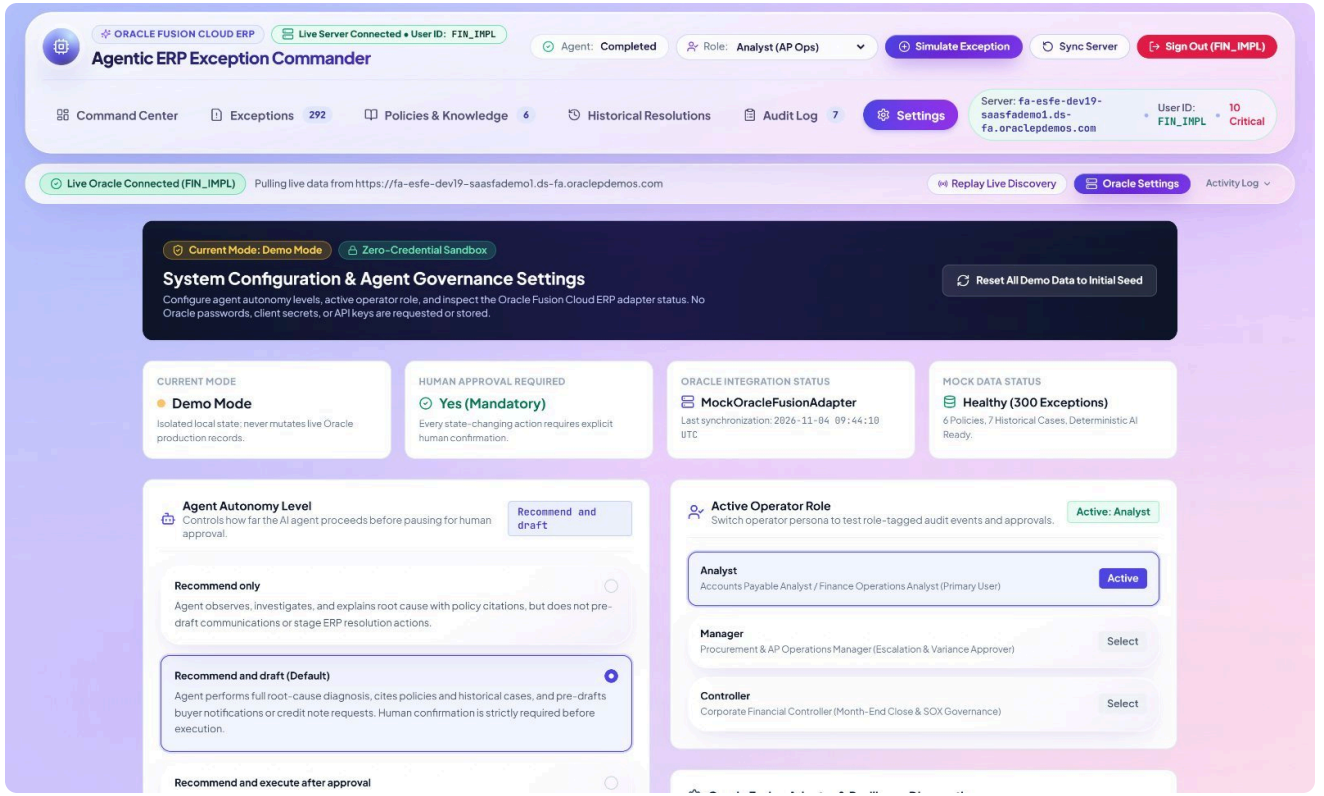
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Configurable Agent Autonomy Tiers & Oracle Server Gateway

Enterprise administrative controls for AI autonomy, role permissions, and ERP connectivity.

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- 3 Autonomy Modes: Advisory Only, Human-in-the-Loop (Recommended Default), and Bounded Auto-Resolve
- Role-Based Access Control switching between AP Analyst, Procurement Buyer, and Controller views
- Live Oracle Fusion REST Endpoint configuration and connection health diagnostics
- Strict Per-User Session Isolation ensuring zero cross-user data leakage



GitHub Repo: https://github.com/Madhuvirat4al/agentic-erp-exception-commander-oracle-fusion

Quantified Business Value & Operational Transformation

Delivering measurable reductions in cycle time, financial risk, and manual effort.

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89% Reduction in Mean Time to Investigate & Resolve (from 38 mins to < 4 mins per invoice)

$310K+ Annual Early-Payment Discounts Protected by accelerating SLA-sensitive approvals

3.2x Faster Month-End AP Subledger Close by proactively clearing period-end blockers

100% SOX Audit Completeness with full human-in-the-loop governance

